

Body-fitted finite-volume NACA0012 verification

Actual body-fitted Gmsh meshes; strict physical relative error <3%.

Scheme: {'momentum_convection': 'linear-upwind', 'pressure_gradient': 'gauss-linear', 'face_interpolation': 'projected-linear', 'wall_gradient': 'linear-upwind'}

CPU calculations; no new GPU or matched TensorLBM result.

Positive mesh geometry and nonorthogonality gate precede all solves.

scale-3: 2190 cells; converged=True; accepted=False

{'drag': 0.12928968858768622, 'lift': 0.21346753343055289}

{'drag_relative_error': 0.0350914, 'lift_relative_error': 0.0627487}

scale-2: 4832 cells; converged=True; accepted=False

{'drag': 0.1278956900099021, 'lift': 0.21557237337687038}

{'drag_relative_error': 0.023931, 'lift_relative_error': 0.0732277}

scale-1: 18712 cells; converged=False; accepted=False

{'drag': 0.12210517325454825, 'lift': 0.19605082725584833}

{'drag_relative_error': 0.0239905, 'lift_relative_error': 0.0249313}

Full independently replayed equations: False

Refinement sequence accepted: False

Asymptotic grid regime qualified: False

Problem, numerical method and reproducibility

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Problem and reference

NACA0012 at $Re=1000$, incidence 4 degrees, unit chord and the reference 36C by 16C domain. The quarter-chord point is (12C,8C). The body is rotated clockwise by 4 degrees about that point, while inlet velocity is (1,0) m/s; drag and lift are global x/y forces in these physical freestream axes. The inlet has uniform velocity; the outlet and open top/bottom boundaries have zero pressure, zero-gradient outflow velocity, and freestream velocity on backflow. The paper specifies open boundaries but does not give an exact FV formula, so this pressure-open implementation is an explicit modeling choice. Geometry/domain values are independently read from [Di Ilio et al. figure 3](<https://arxiv.org/html/2006.10487>). Figures 10 and 11 give digitized coefficient references and ± 0.0001 graph-reading intervals; worst endpoint errors are used. The domain and boundary review is archived in docs/naca-reference-domain-review.json.

Mesh and discretization

Gmsh 4.15.2 frontal Delaunay generates conforming triangles with physical boundary tags; cylinder CAD arcs or a closed NACA0012 interpolating spline define the body. Physical wall faces are linear chords. See the [Gmsh official manual](<https://gmsh.info/doc/texinfo/gmsh.html>) for algorithm 6 and Distance/Threshold fields. Neither Cartesian obstacle masks nor staircase boundaries are used. Positive volumes and projected face distances and a maximum nonorthogonality below 70 degrees are required before solving. The 70-degree project gate is a screening criterion, not proof of sufficient accuracy.

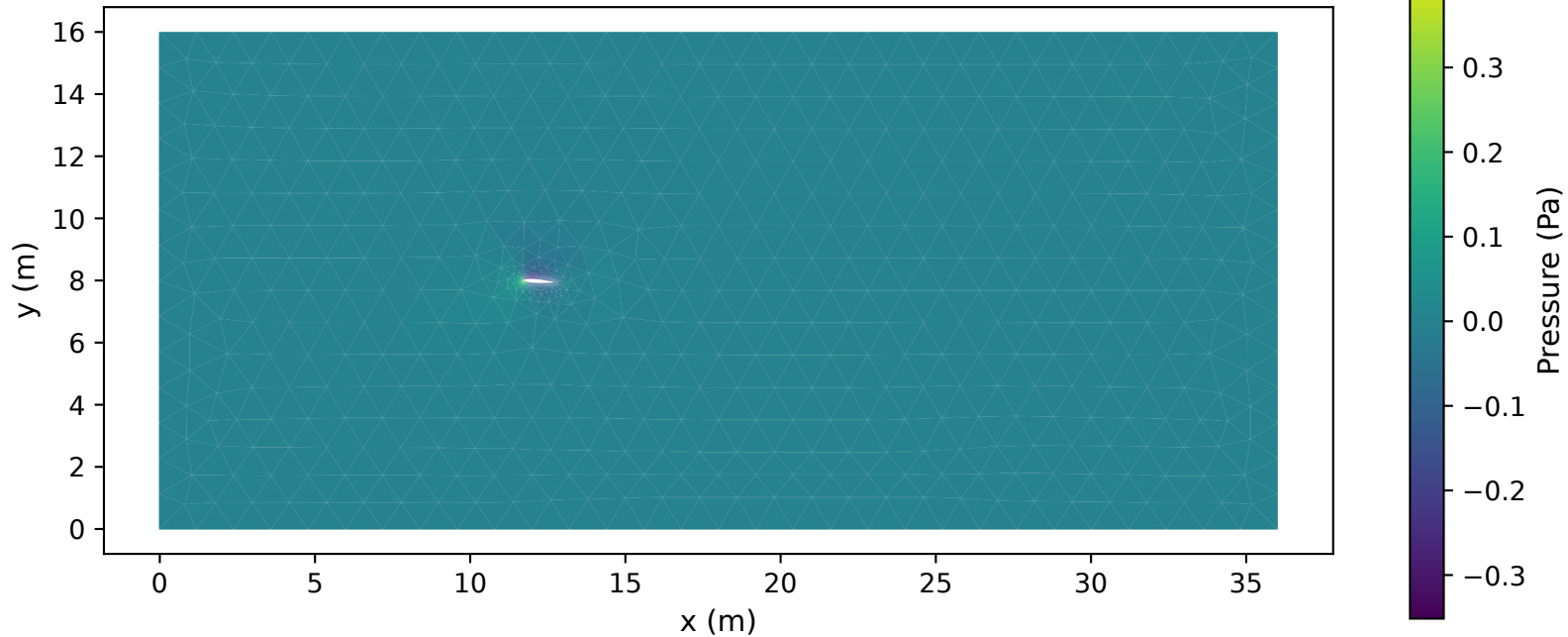
The recorded discretization is ``{"boundary": "pressure-open-freestream-backflow", "face_interpolation": "projected-linear", "momentum_convection": "linear-upwind", "pressure_gradient": "gauss-linear", "wall_gradient": "displacement-correction"}``. Linear upwind uses the least-squares gradient of the upstream cell and the actual face-to-cell displacement; its conservative deferred correction is inspired by [OpenFOAM linearUpwind](<https://github.com/OpenFOAM/OpenFOAM-dev/blob/master/src/finiteVolume/interpolation/surfaceInterpolation/schemes/linearUpwind/linearUpwind.C>). This is unlimited reconstruction, so no bounded/TVD claim is made. Gauss pressure gradients use geometric face interpolation and the same boundary face pressure used by pressure traction. Nonorthogonal diffusion, physical Rhie-Chow flux and no-slip reconstructed molecular stress are retained. Cylinder acceleration uses a full-grid coupled Picard matrix with an independent response check; NACA uses SIMPLE and Anderson mixing. CPU sparse solves check true residuals. No reference value enters the equations, initialization or relaxation.

Reproduction and acceptance

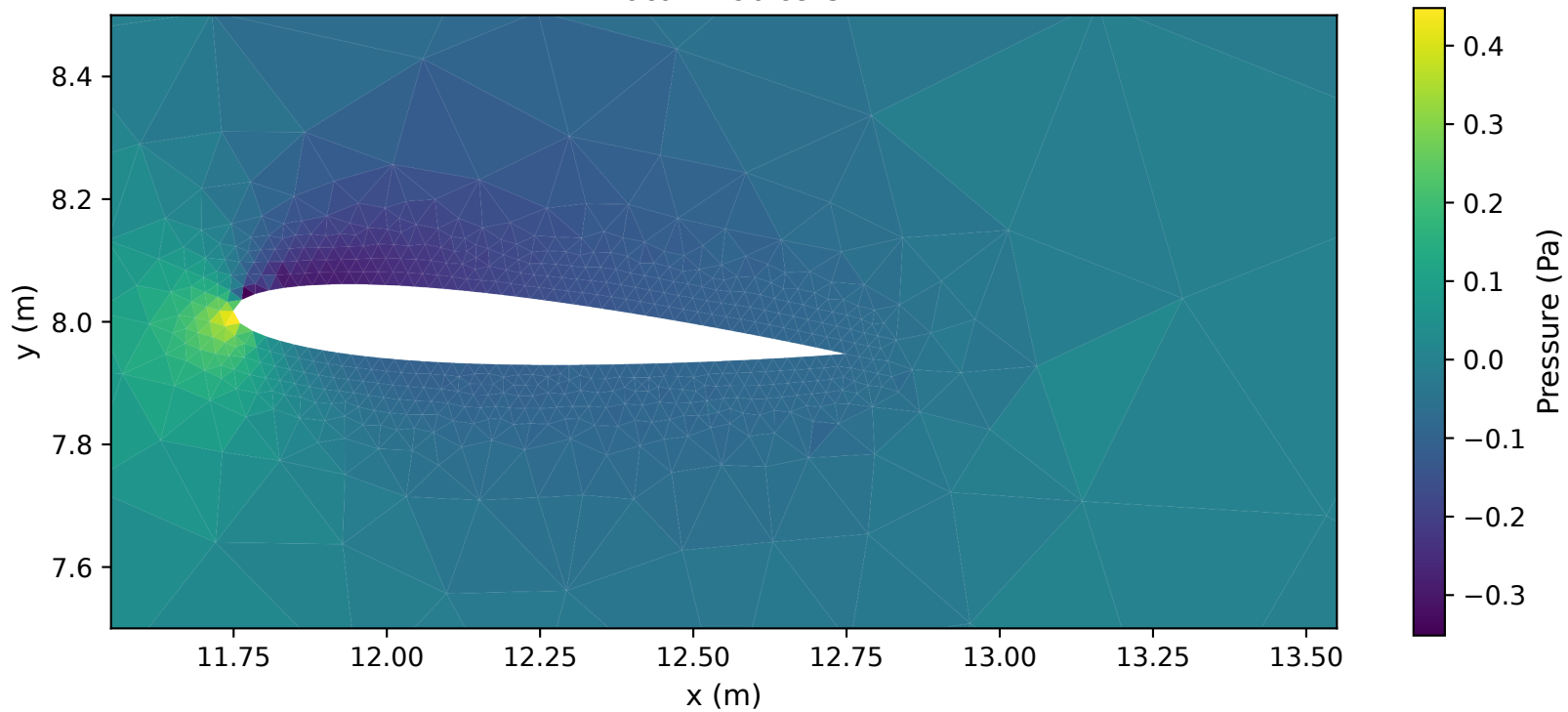
```
```bash
uv pip install --python /path/to/python gmsh numpy scipy matplotlib
OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 PYTHONPATH=src python -m
tensorfvm.verification.external_matched_naca --scales 3.0 2.0 1.0 --max-iterations 2500 --output
results/naca
PYTHONPATH=src python scripts/audit_external_equations.py results/naca
PYTHONPATH=src python -m tensorfvm.verification.external_mesh_report results/naca
```
```

Raw mesh/field files, velocity/pressure arrays, face mass fluxes, separate tractions, full histories, linear residuals and SHA256 source/artifact manifests accompany this report. Each physical relative error must be strictly below 3%; steady equation residuals must be below $1e-6$ and the physical flux fixed-point defect below $1e-7$. Source-preserving independent NumPy replay checks full configured equations. Three runs do not establish an asymptotic grid regime; qualification of the finest run does not qualify the entire refinement sequence. GPU and matched TensorLBM runs have not been performed for this new unstructured path.

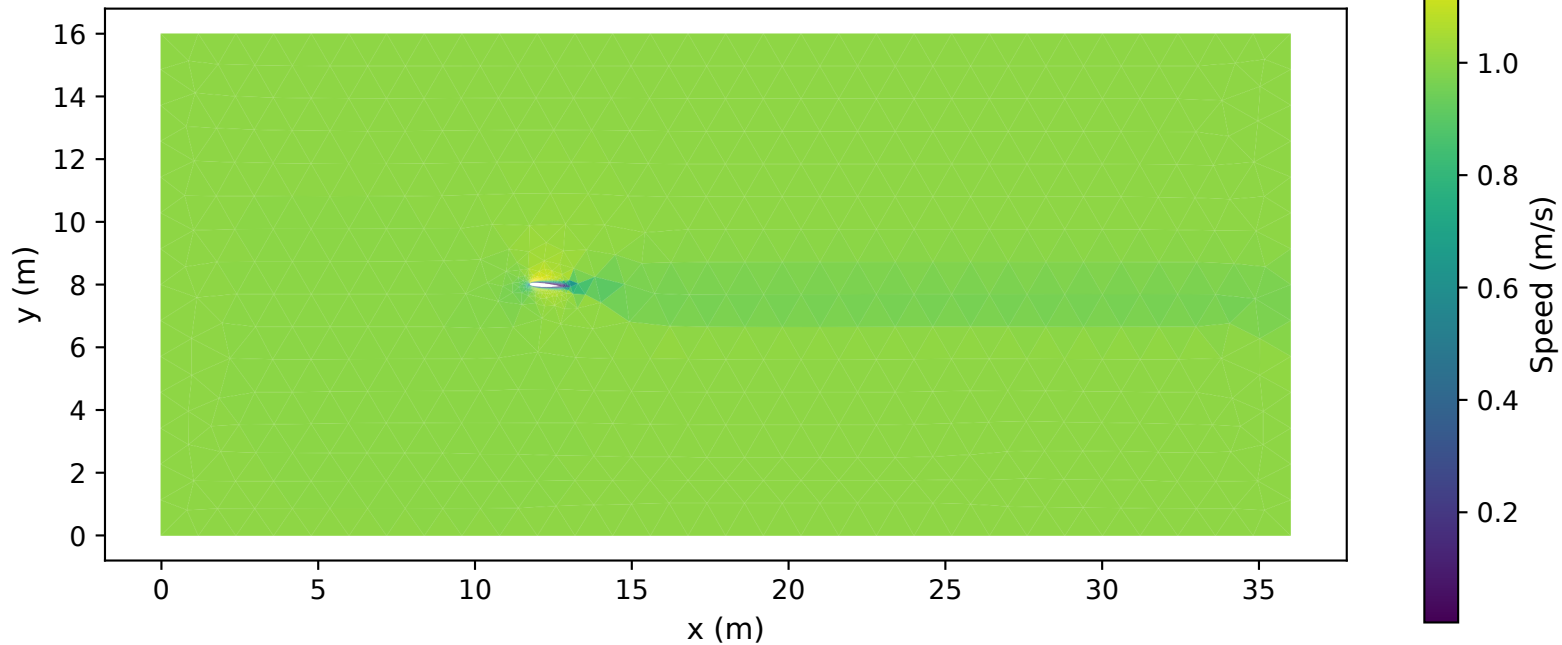
naca: 2190 cells



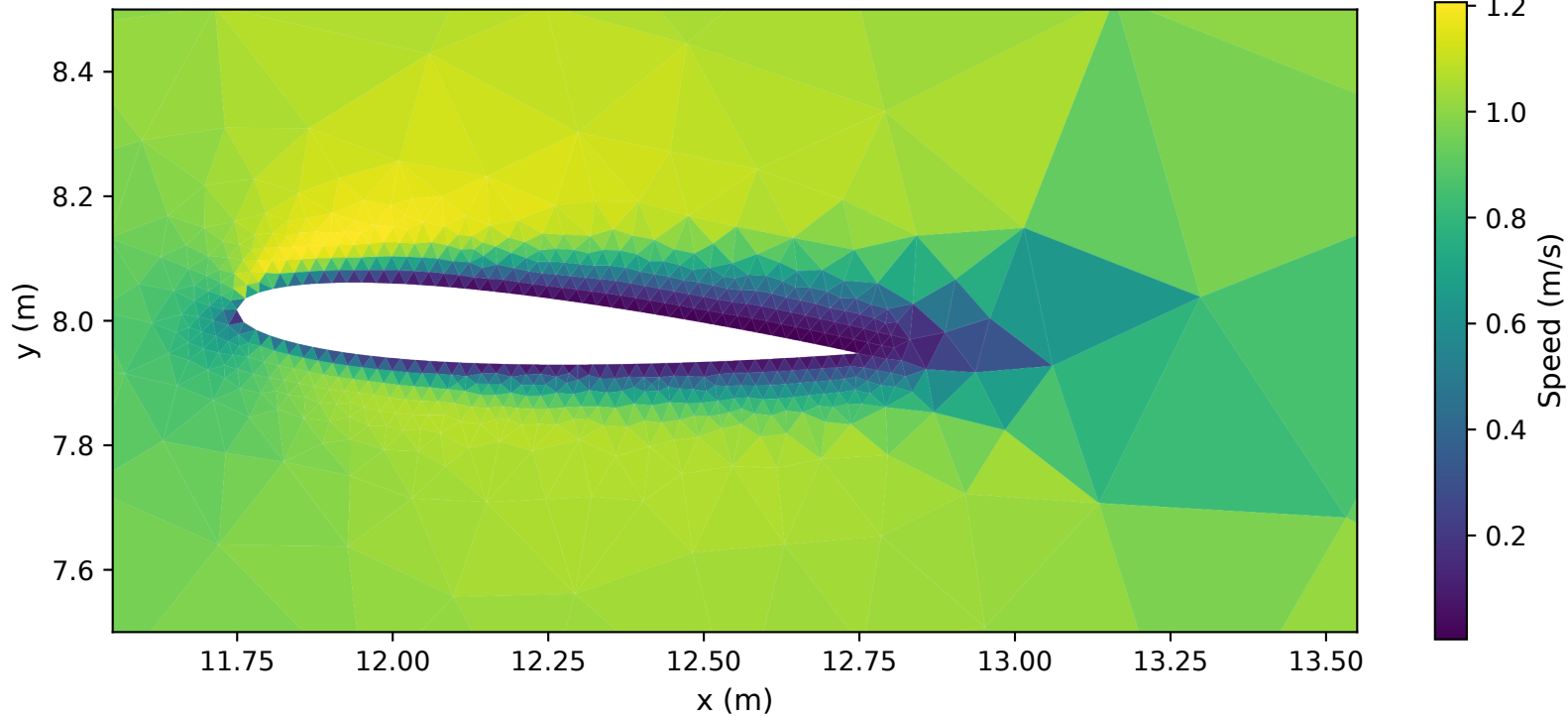
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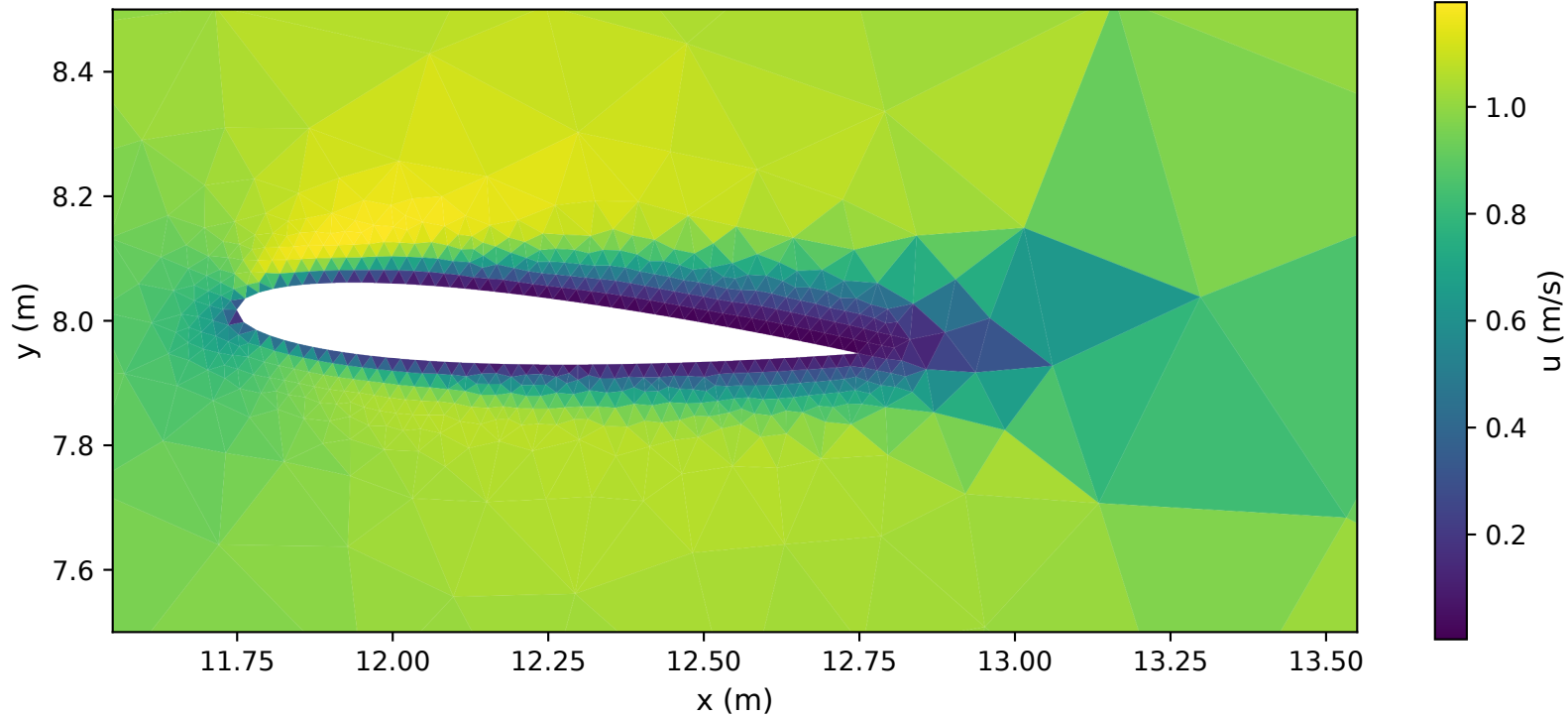
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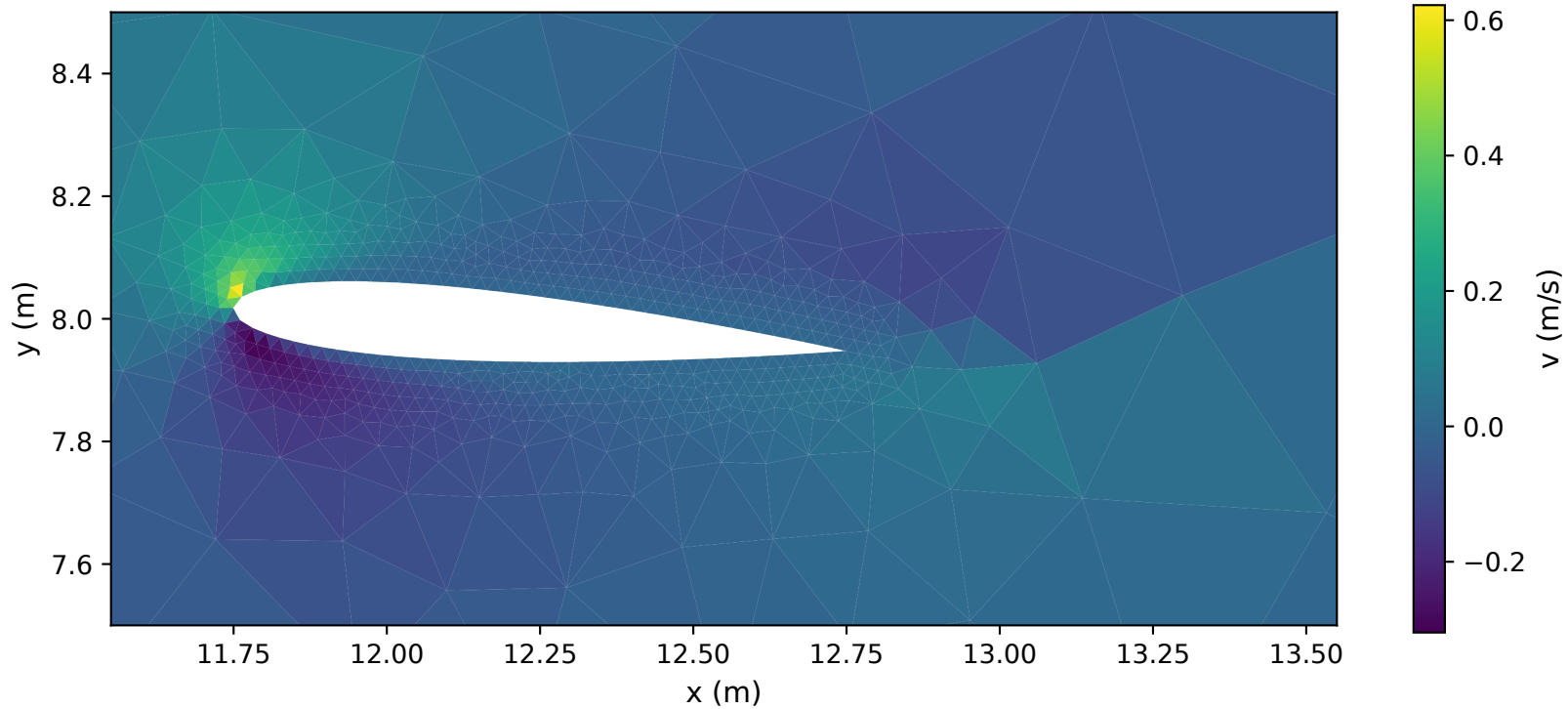
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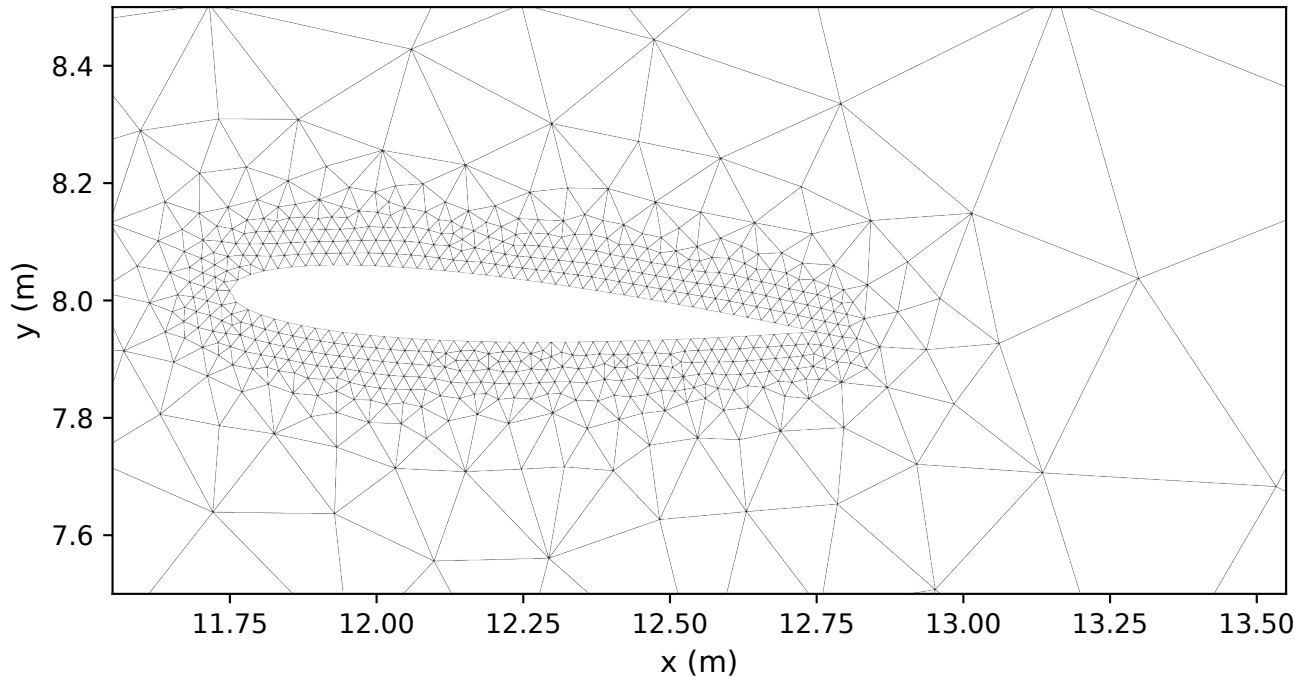
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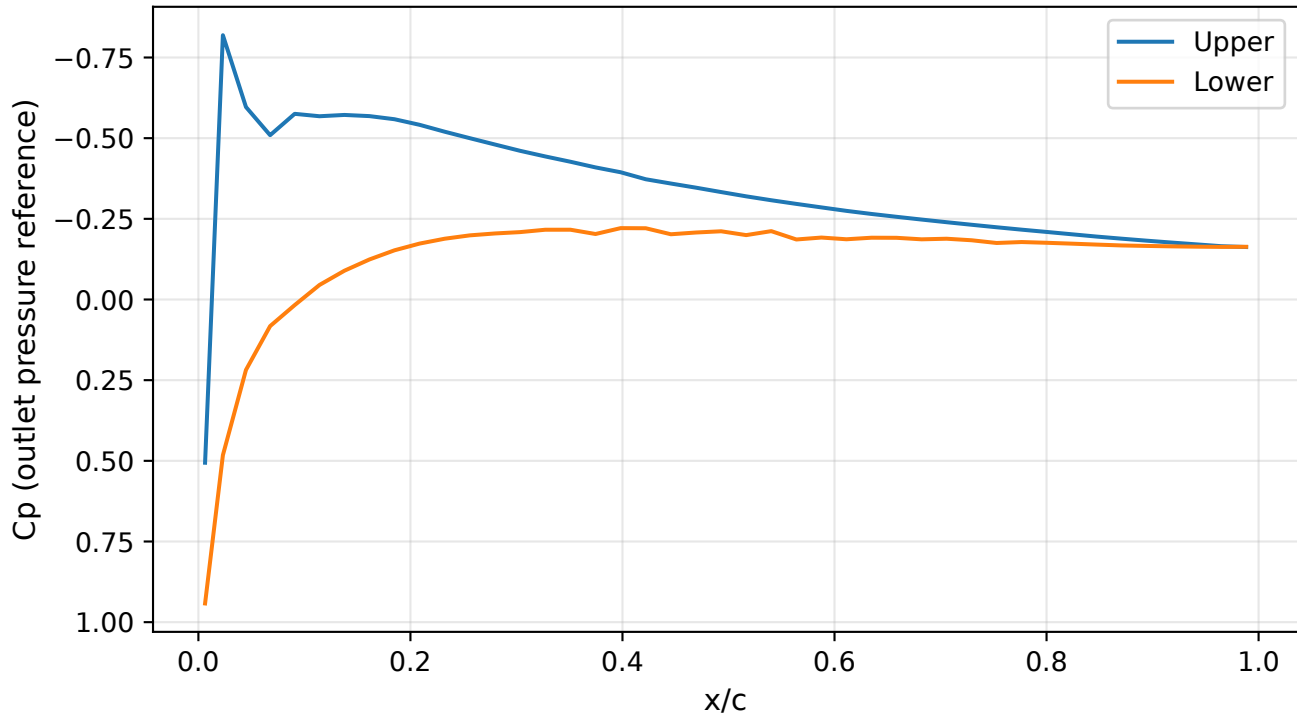


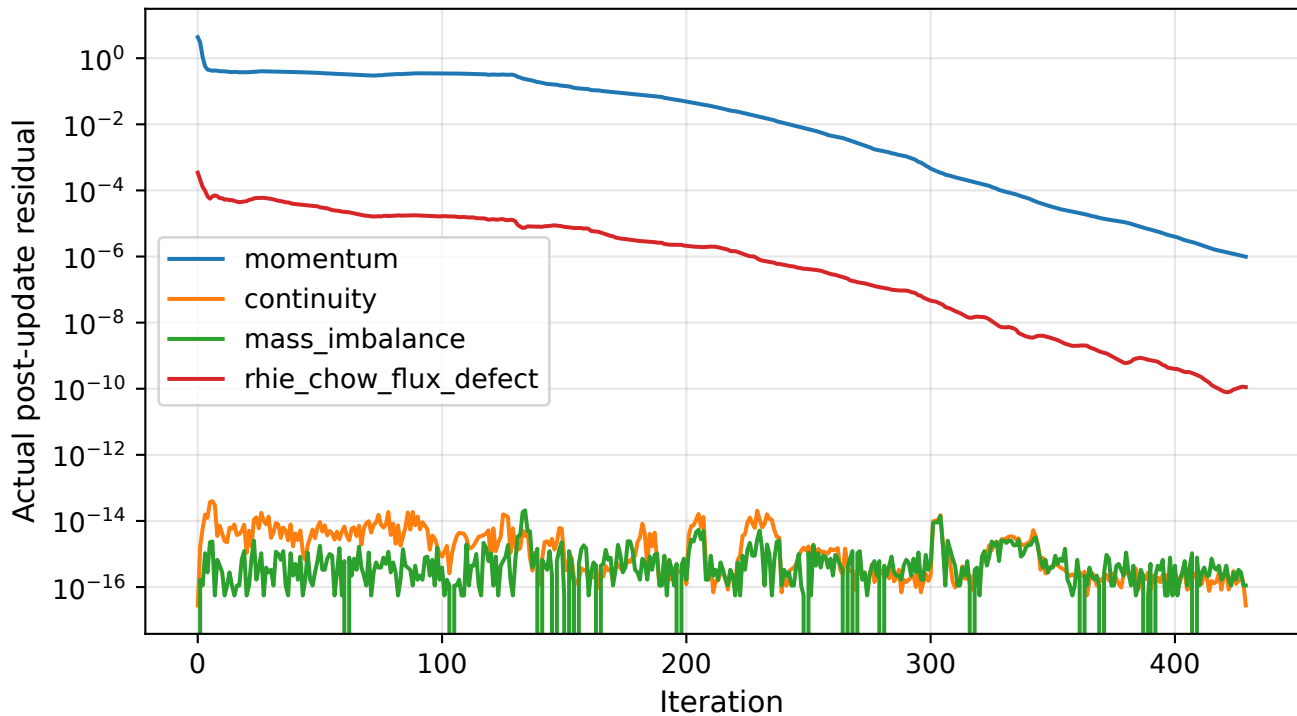
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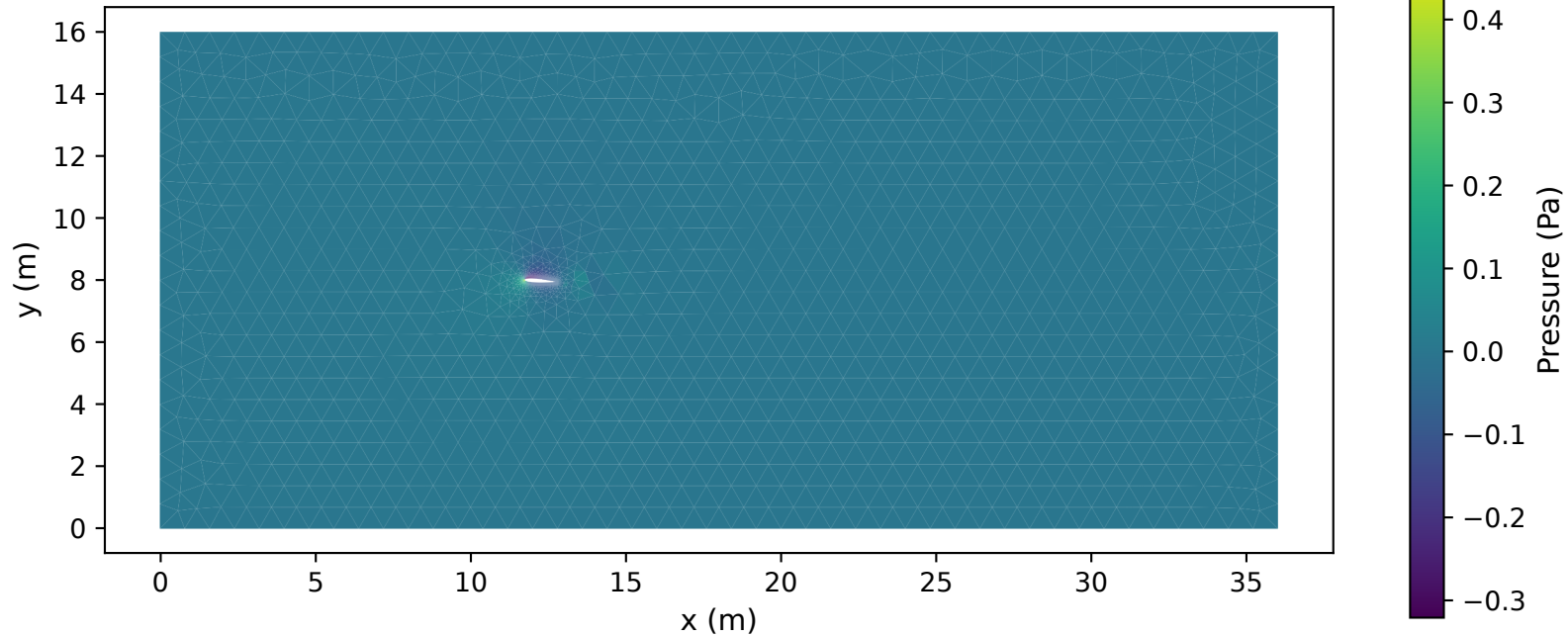
Actual body-fitted mesh



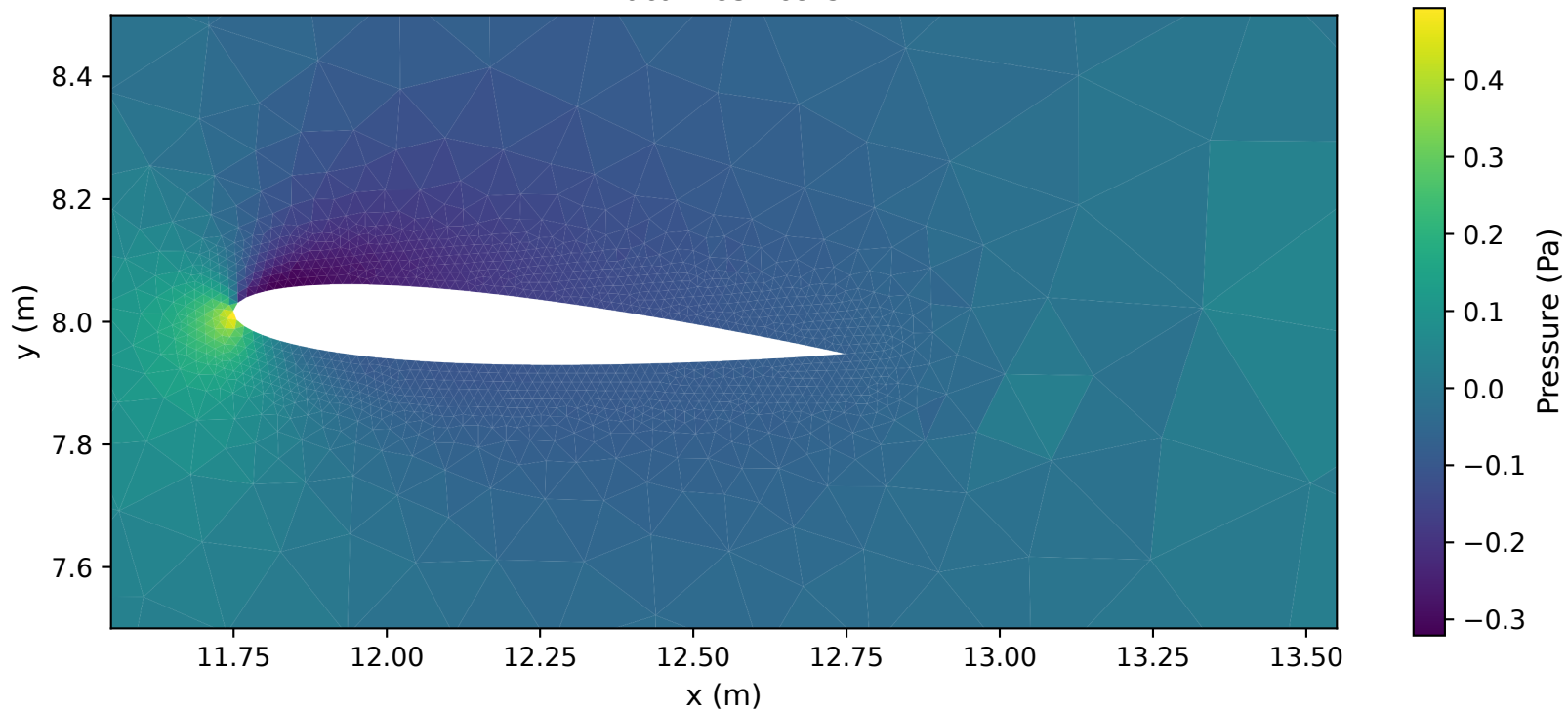




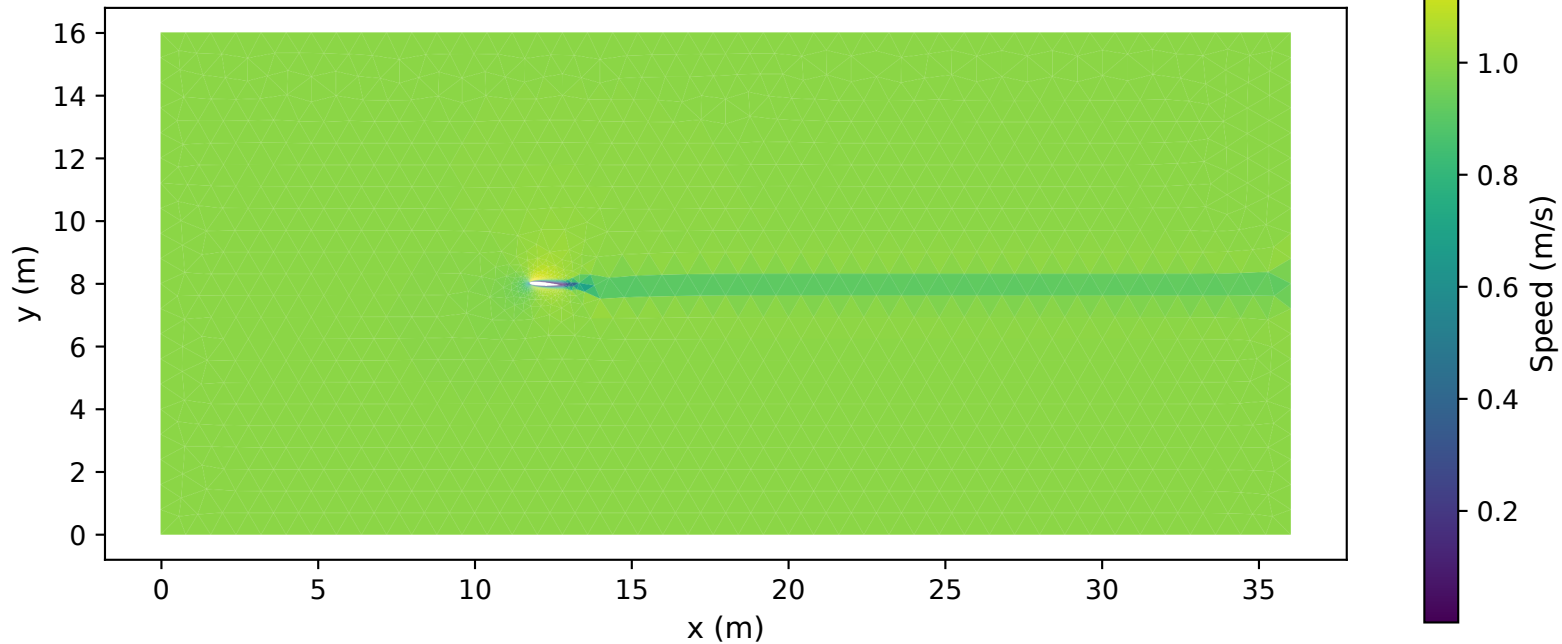
naca: 4832 cells



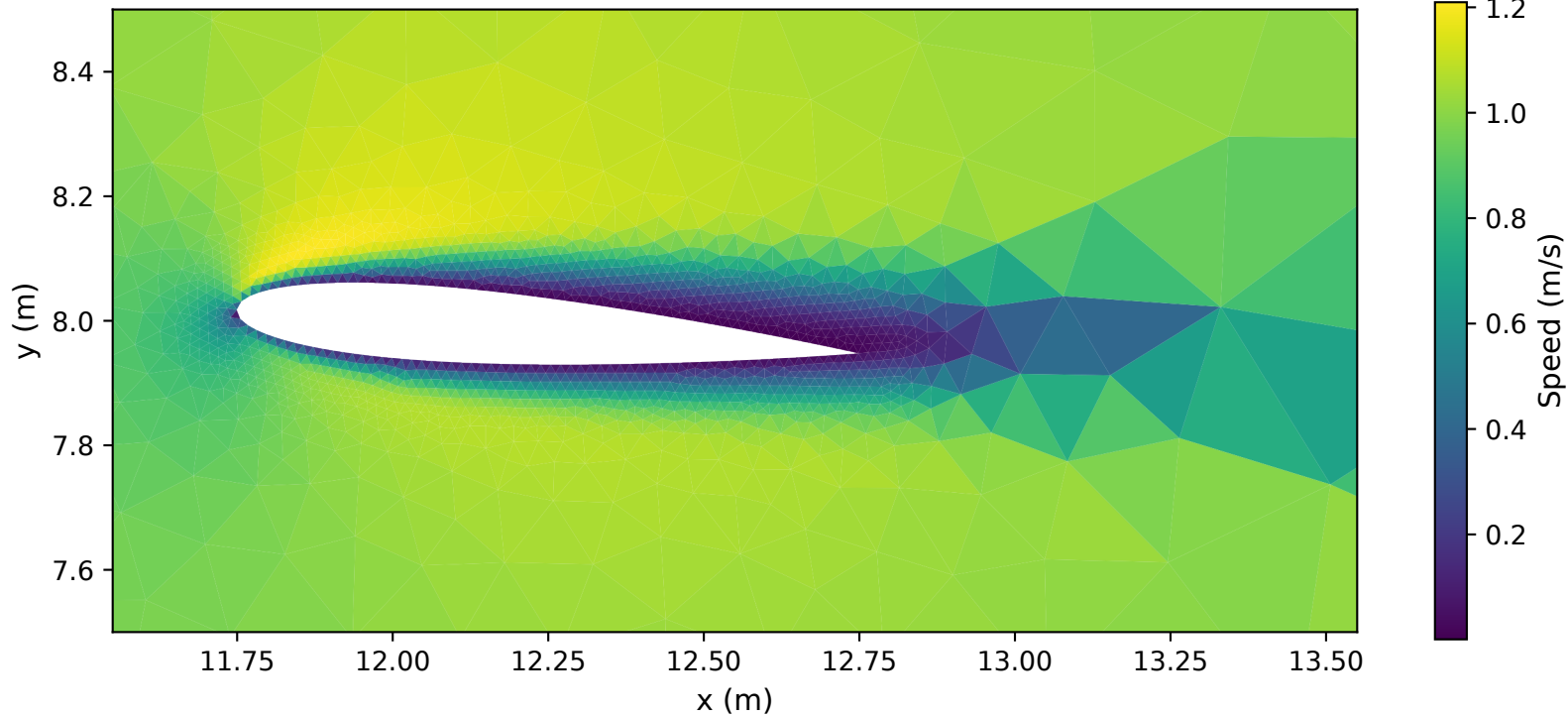
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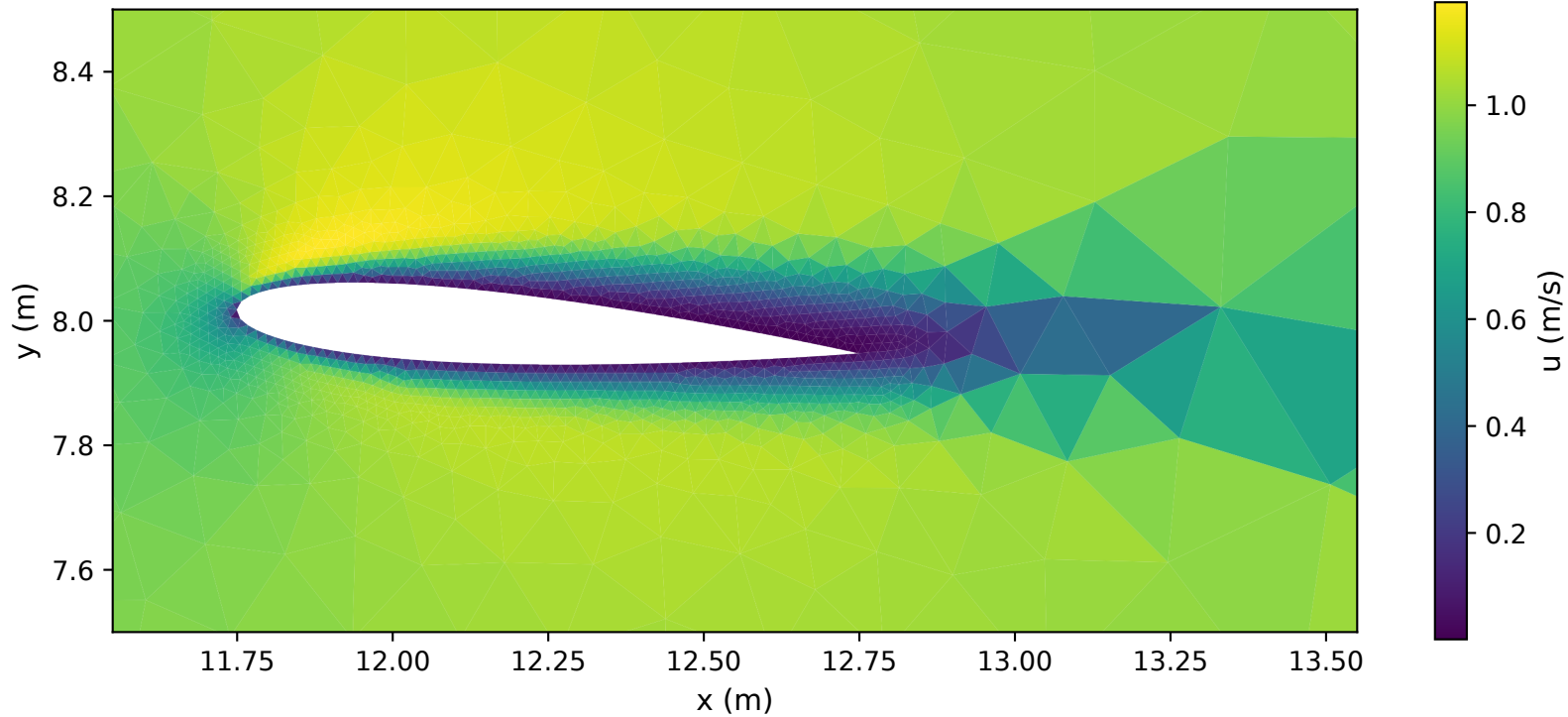
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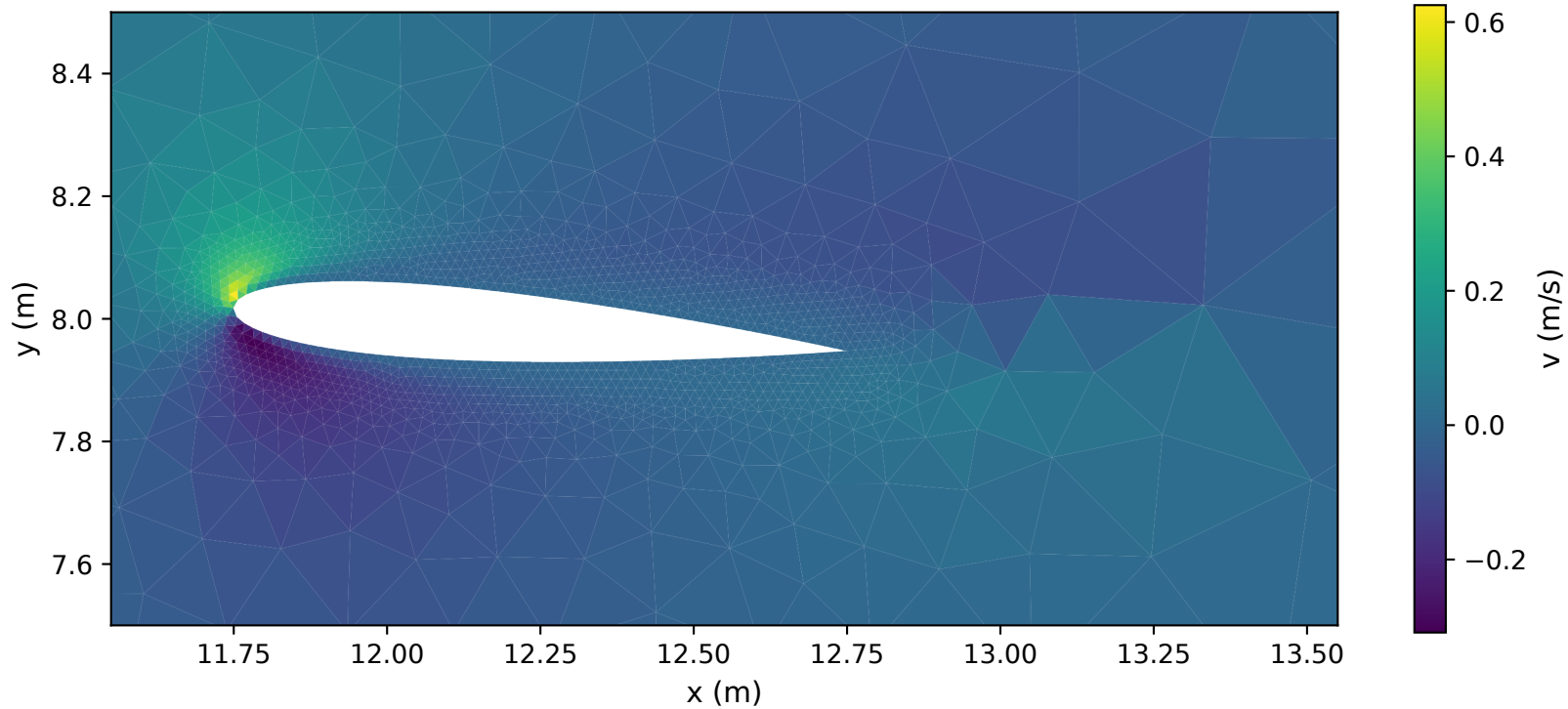
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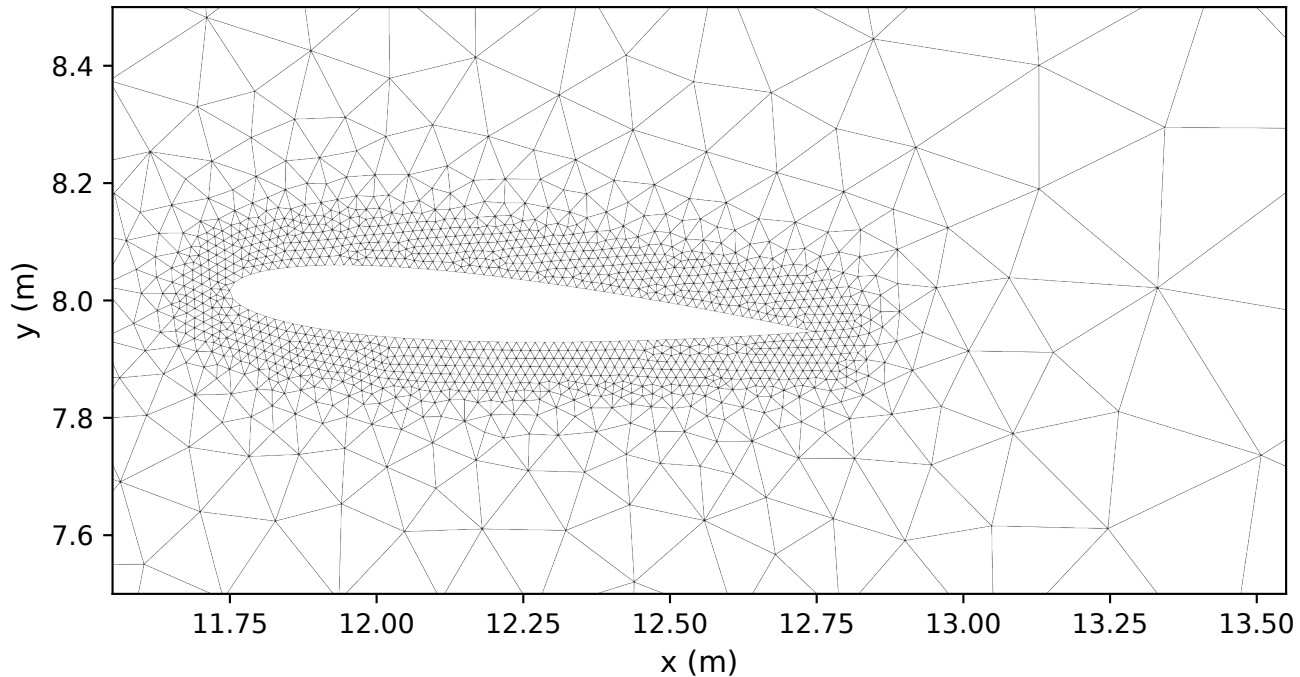
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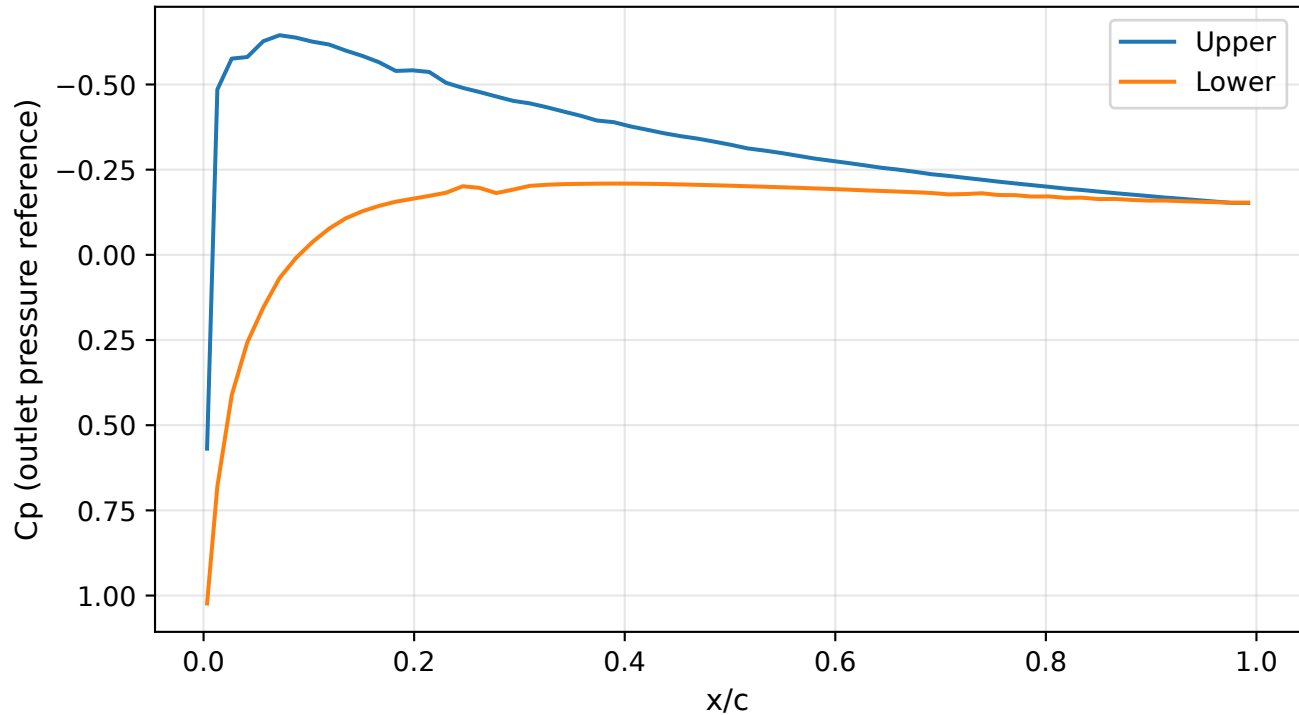


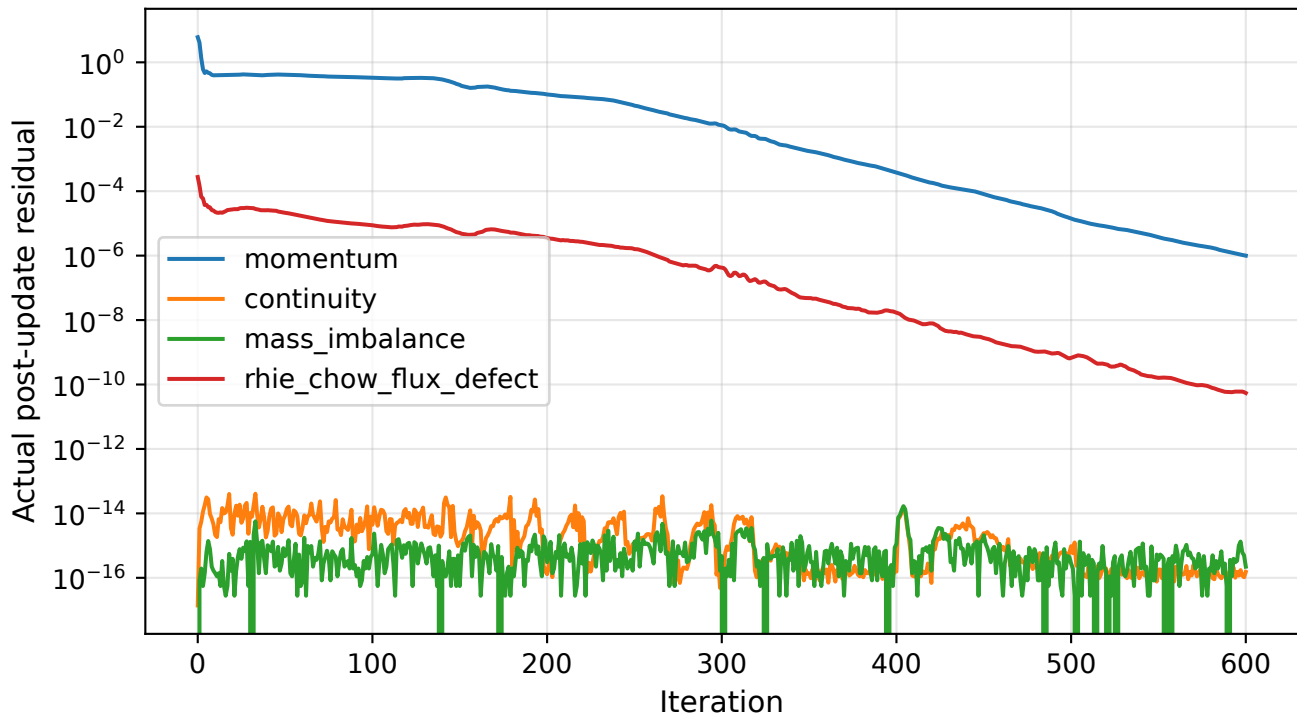
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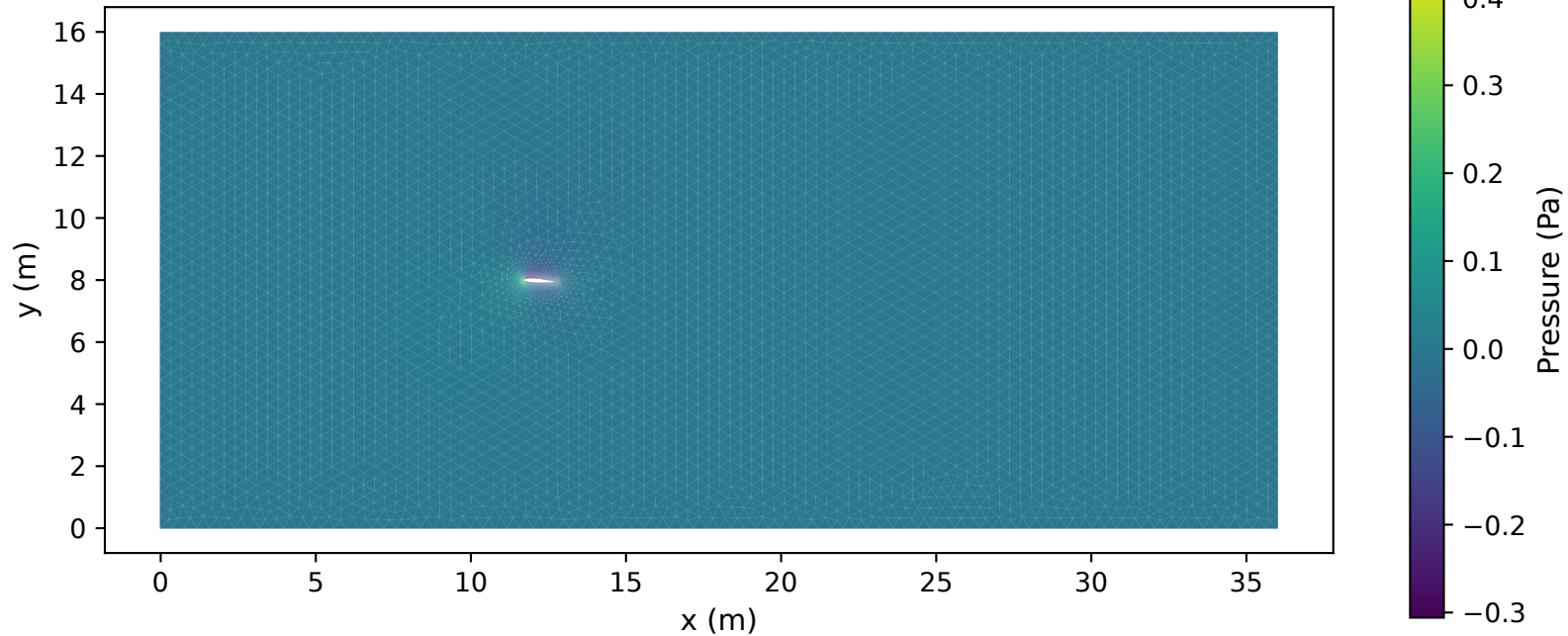
Actual body-fitted mesh



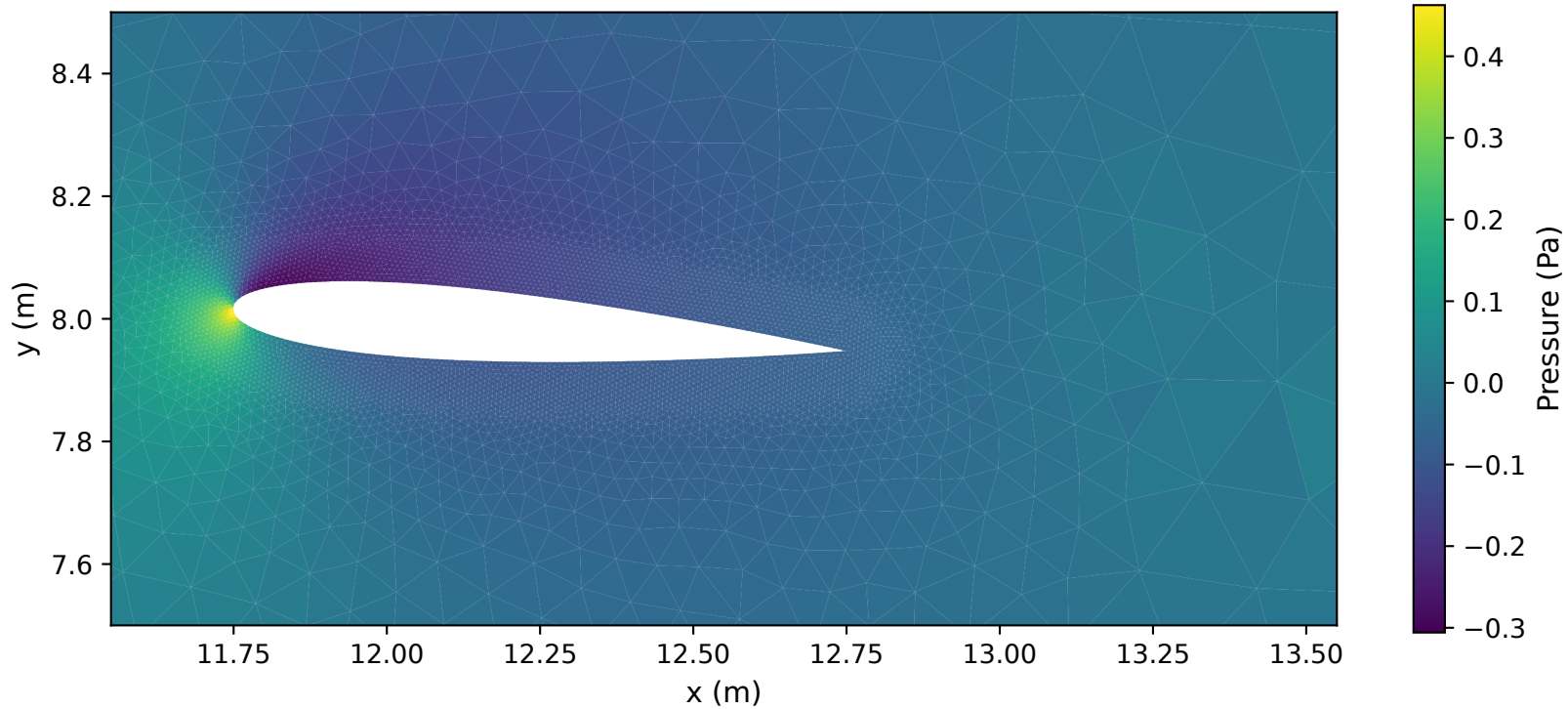




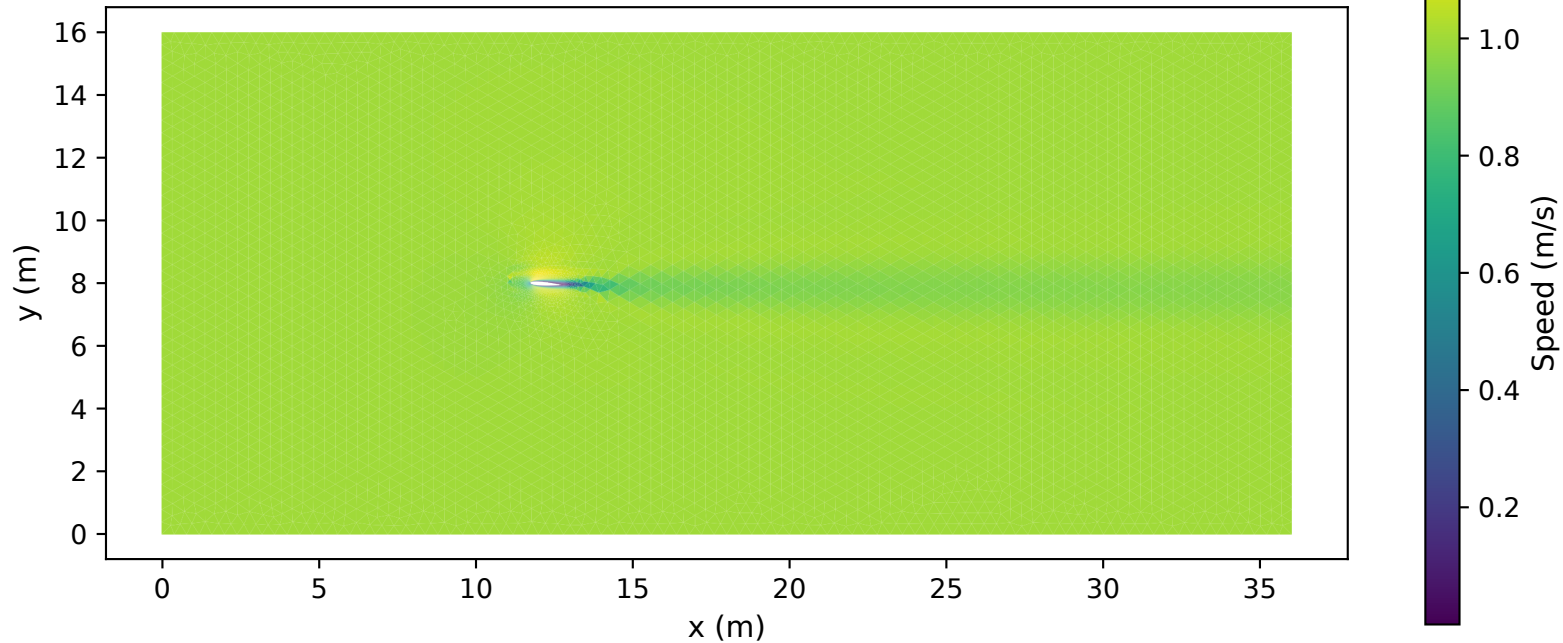
naca: 18712 cells



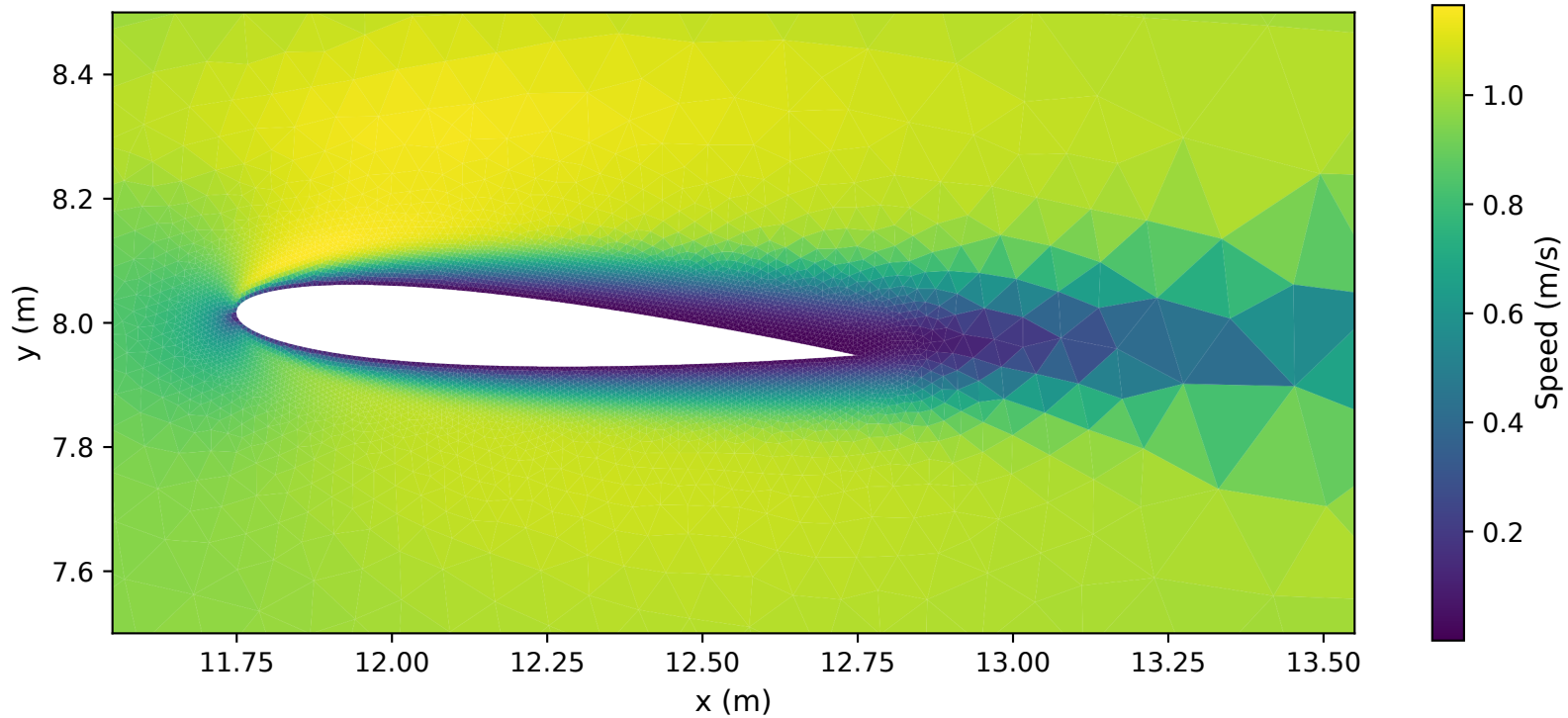
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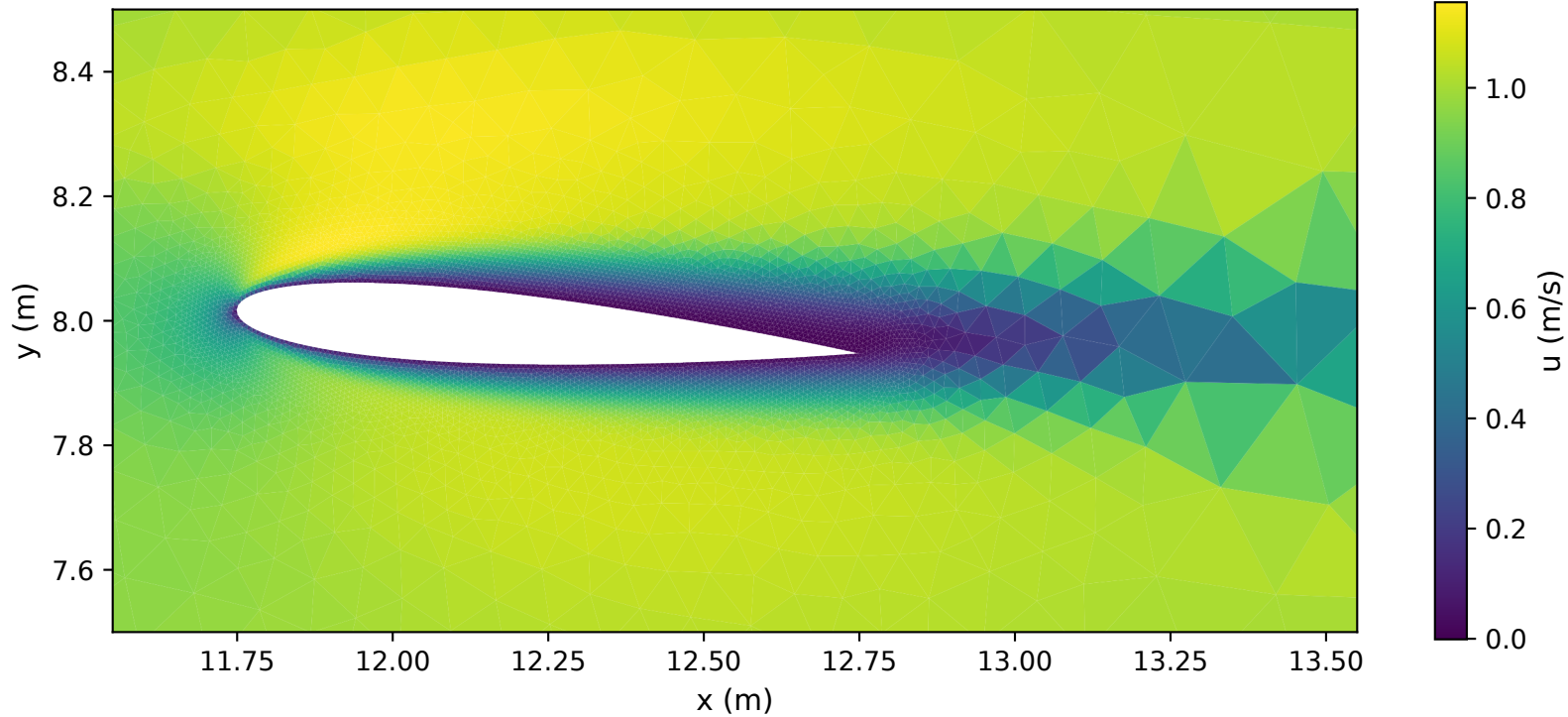
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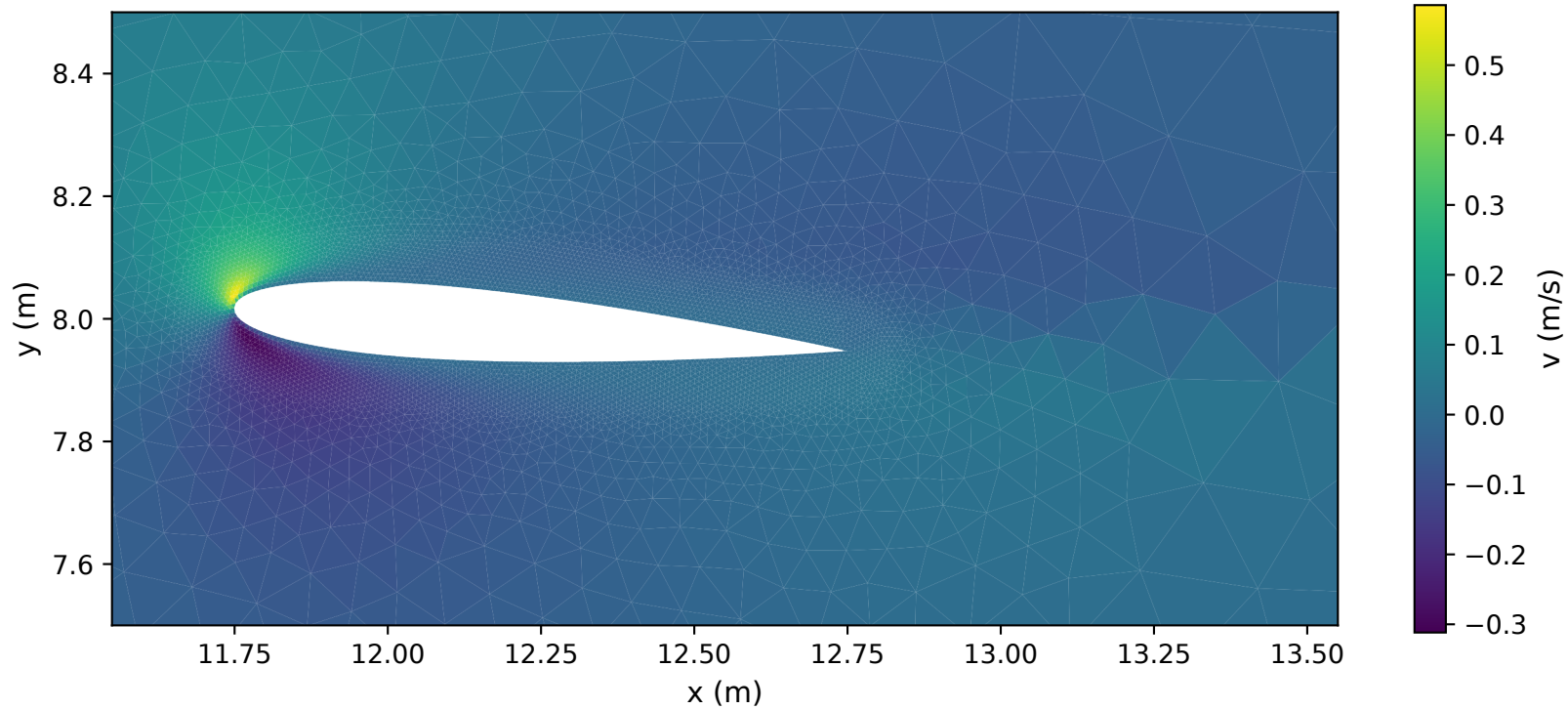
naca: 18712 cells



naca: 18712 cells



naca: 18712 cells



Actual body-fitted mesh

